

Kareem H. El-Safty

Quantum Software
Engineer — QML

On site or remote - Available immediately - Former Qiskit Advocate

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Professional Summary

Quantum software engineer with a photonics backbone. I build power-aware MZI pipelines and CV/Kerr kernels, ship open-source tools (PhotonWeave, KTQ; contributor to Piquasso), and take experiments from notebook to AWS Braket. Ex AI team lead with production ML wins (face verification, anti-spoofing, OCR, industrial vision), former Qiskit Advocate, and active mentor/community builder (QEgypt, Alexandria QCG).

Skills

Programming Python (advanced), C++ (beginner–intermediate), Git, Docker, Linux, CI/CD, VS Code.

ML/Math scikit-learn (expert SVM/trees/pipelines), TensorFlow, JAX, Keras/TensorFlow, Optuna, AutoKeras, KerasTuner, custom losses/gradients.

Quantum Qiskit, PennyLane, Cirq, Strawberry Fields, Perceval, The Walrus, Piquasso, PhotonWeave, AWS SDKs Braket.

Photonics/QML MZI calibration and energy optimization; CV and Kerr kernels; joint detection receivers; photonic simulation.

Numerics einsum/tensordot patterns; JAX and Numba acceleration; experiment orchestration; reproducible pipelines.

Cloud AWS EC2 and Braket, GCP.

Productivity MS Office (Word, Excel, PowerPoint), Google Workspace (Sheets, Slides), LaTeX.

Languages English C2 - Arabic C2 - German A1

Employment

- Nov 2022 – Jul 2025 **Researcher, Quantum Machine Learning, Technical University of Munich (TUM), Germany**
- Calibrated reconfigurable **MZI photonic processors** with **energy optimization**; improved transform fidelity while minimizing heater power (see Optics Express 2025).
 - Designed and benchmarked **CV/Kerr kernels**; integrated with JAX/NumPy for fast sweeping and analysis.
 - Explored **joint detection receivers** and energy-aware quantum communication; collaborated across physics and platform teams.
 - Built **reproducible research stacks** (tests, docs, CI) and CLI tooling; **mentored two MSc** students end-to-end.
- Apr 2020 – May 2022 **Research Assistant (Quantum Computing), Wigner Research Centre for Physics, Budapest, Hungary**
- Continuous-variable photonic simulation and QML experiments; contributor to the **Piquasso** simulator and documentation.

- Jul 2018 – **AI Team Leader**, *DevisionX*, Cairo, Egypt
- Nov 2021
- Delivered ML systems **directly with business clients**: scoped requirements, set milestones, and deployed to embedded/DSP and production.
 - Projects: **face verification and anti-spoofing**, **Arabic OCR**, and industrial machine vision pipelines.
 - Implemented **AutoML** (Optuna/AutoKeras/KerasTuner) and custom loss functions; accelerated iteration and robustness.
 - Ran customer discovery and requirements workshops; wrote PRDs and delivery plans; managed milestones to production.
 - Partnered with sales on tenders, demos, and acceptance criteria; maintained a living roadmap and metrics.
 - Built and led the **intern program** (screening → onboarding → training).
- Aug 2021 – **Machine Learning Instructor (Freelance)**, *Information Technology Institute (ITI)*, Egypt
- Present
- Taught supervised/unsupervised ML; designed labs and assessments; maintained calendars and materials.

Education

- 2021 – 2022 **M.Sc. — Quantum Computing Technologies**, *Universidad Politecnica de Madrid*, Spain
Thesis: **Quantum continuous variable kernels for QML**.
- 2019 – 2020 **Pre-Master (1.5 years)**, *Bridging Communications and Electronics with Computer Engineering*, Egypt
Coursework and projects **bridging the three tracks**.
- 2012 – 2017 **B.Sc. — Electrical, Electronics and Communications Engineering**, *Suez Canal University*, Egypt
Thesis: **Humanoid robot using voice and object recognition**.

Selected Projects

- MZI power-aware synthesis Developed an algorithm to reduce photonic chip power while realizing arbitrary unitaries with MZI decompositions; integrated into calibration workflows.
- Braket Nash workflows End-to-end cloud execution on **AWS Braket** for game-theoretic optimization (Nash equilibrium); cost/perf trade-offs and result handling into analysis notebooks.
- Industrial vision Built face verification and **anti-spoofing**, **Arabic OCR**, and inspection pipelines; delivered against client milestones; handover with documentation.
- AutoML stacks Orchestrated Optuna/AutoKeras/KerasTuner pipelines; designed custom loss/metrics; reproducible CI-first model iteration.

Selected Publications

- 2025 **El-Safty K.H.** et al. Robust calibration and energy optimization in reconfigurable photonic processors. *Optics Express* 33(16):35011–35027.
- 2025 Kolarovszki R. et al., **El-Safty K.H.**. Piquasso: A Photonic Quantum Computer Simulation Software Platform. *Quantum* 9:1708.
- 2025 Sekavcnik T.; **El-Safty K.H.**; Nötzel J. PhotonWeave. *Journal of Open Source Software*.
- 2025 Khan F.S.; Okrut O.; Cannon K.; **El-Safty K.H.**; Elsokkary N. Calculating Nash equilibrium on quantum annealers. *Annals of Operations Research* 346(2):1109–1126. doi:10.1007/s10479-023-05700-z
- 2024 Amiri H.; Dehdashti S.; **El-Safty K.H.**; Litvin L.; Munar-Vallespir D.; Nötzel J.; Sekavcnik T. Quantum advantages for data transmission in future networks. *Computer Networks* 254:110727.
Full list: Google Scholar.

Research Software

- PhotonWeave Photonic toolkit with tests, docs, CI, and CLIs. github.com/kareem1925

KTQ Modular QML and photonic research utilities (NumPy/JAX). github.com/kareem1925
Piquasso Continuous-variable photonic simulator. github.com/Budapest-Quantum-Computing-Group/piquasso
(contributor)
QuReed Quantum simulation tooling (preprint). [arXiv:2406.07638](https://arxiv.org/abs/2406.07638)

Community and Outreach

QEgypt Co-founder, national quantum-education chapter (events, workshops, mentoring). qworld.net/qegypt
Alexandria Organizer and past content lead (SciAlex); manage the group **YouTube** channel. SciAlex |
QCG YouTube
Teaching Instructor, QBronze/QSilver/QNickel; mentor of two MSc students (QML/photonic topics).
Service Reviewer: *New Journal of Physics* (IOP), *SN Computer Science* (Springer). Program committees: Geekle (Python for ML and AI), ICCS Quantum Computing Track.
Advocacy Former **Qiskit Advocate**; community talks and workshops on QML and photonics.

Certificates and Courses

AWS Certified Machine Learning — Specialty (Credly).
Udacity AI Nanodegree (Credly).
Stanford Online — Quantum Mechanics for Scientists and Engineers 1 & 2.
DelftX — Quantum Cryptography.
TorontoX — Quantum Machine Learning.
Coursera — Deep Learning Specialization.

Additional Details

Focus Areas QAOA/variational circuits; CV/Gaussian and non-Gaussian models; kernel theory and quantum-inspired kernels; annealing embeddings.
Tools VS Code, LaTeX; data handling with Pandas; plotting with Matplotlib; experiment tracking in notebooks and Git.